WPR - 9

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY
B.Tech (CSE - AIML) VII Semester

Project Area -- Generative AI & AI Safety (Trustworthy AI / MLOps)

Students Weekly Progress Report (WPR — Week 9) For Odd Semester of session 2026-2027

To be filled by Students	Group No. 50
Students Name	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh
Roll no.	1. 569 2. 166 3. 191
Enrollment no.	1. A2305223569 2. A2305223166 3. A2305223191
Project Title finalized, if Yes, give name, if NO, give reason	Yes. Title: "GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense"
Synopsis submitted	Yes (Approved for Group No. 50)
Literature review	Expanded to premier published research corpora: Stanford SPML Chatbot Injections (16K), Jayavibhav Prompt Injections (327K), Deepset Benchmark, ChatGPT JailbreakBench DAN Personas, FEVER NAACL 2018, and HaluEval EMNLP 2023 (35K).
Technical & Economical Feasibility	Feasible. Local NVIDIA GeForce RTX 3050 GPU (4GB VRAM), PyTorch 2.6 CUDA, Hugging Face Transformers, DeBERTa-v3 SLM (86M). Zero monetary cost.
Bill of Material	NA
Project Progress Schedule (PERT Chart)	• W1-W5: Setup, dataset curation, DeBERTa fine-tuning & evaluation (Completed) • W6-W8: Streaming entropy engine, NLI grounding & comparative benchmarking (Completed) • W9: Massive multi-dataset scaling (5,796 samples), Stanford SPML/Jayavibhav/HaluEval ingestion, local GPU fine-tuning, in-distribution & zero-shot OOD cross-benchmark validation (Completed / Week 9) • W10-W12: IEEE research paper manuscript drafting, ablation experiments, and final project viva defense preparation.
Design of critical components	1. Mega Benchmark Ingestion & Stratified Splitting Pipeline (`src/prepare_splits_v3_mega.py`) 2. Stage-1 DeBERTa-v3 High-Capacity GPU Training Engine (`src/train_prescan_v3.py`) 3. Multi-Benchmark Evaluation & Cross-Domain OOD Testing Suite (`src/evaluate_full_suite_v3.py`) 4. 3-Way Benchmark Progression Table & Dual Confusion Matrix Heatmaps
Fabrication work (give %)	0% (Software Project)

Experimental work (give %)	90% (Multi-benchmark scaling completed across 5,796 samples; established 98.71% in-distribution accuracy and 98.20% zero-shot OOD generalization)
Result and Analysis	Evaluated on 698 held-out test samples and 500 zero-shot OOD samples. Intercepts 98.03% of in-distribution attacks and 97.20% of novel jailbreaks with sub-11ms latency on RTX 3050 GPU; false positive rate suppressed to 0.58%. Stage 2 fact grounding intercepts 78.25% of hallucinations with 16.45ms latency.
Report writing	WPR Week 9 drafted.
Signature of students	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh

Work done in this week (Week 9):

1. Massive Multi-Dataset Ingestion (5,796 Total Samples): Following mentor guidance for upcoming research paper submission, expanded benchmarking from initial 451/1,182 samples to premier published research corpora: Stanford SPML Chatbot Injections (16K corpus), Jayavibhav Benchmark (327K corpus), Deepset Benchmark, ChatGPT JailbreakBench DAN Personas, FEVER NAACL 2018, and HaluEval EMNLP 2023.
2. Stratified Splitting & Out-of-Distribution (OOD) Pipeline: Partitioned the 5,000-sample security corpus into 70% Train (3,253 samples), 15% Validation (697 samples), and 15% Test (698 samples). Engineered a separate 500-sample zero-shot OOD test set featuring novel jailbreak vectors (DAN roleplay, base64 obfuscations, indirect injections) to rigorously test cross-dataset generalization.
3. Local GPU Model Fine-Tuning: Successfully fine-tuned `microsoft/deberta-v3-small` (86M parameters) across 3 epochs (612 steps) on local NVIDIA GeForce RTX 3050 GPU, achieving 98.28% validation accuracy and 0.9826 F1-score with 0.1024 validation loss.
4. Empirical Multi-Benchmark Results: Evaluated across held-out and OOD test splits, establishing:
   • In-Distribution Test (698 samples): 98.71% Accuracy, 98.03% Attack Recall (catches 348 of 355 attacks), 99.43% Precision, 0.58% False Positive Rate, and 10.10 ms median GPU latency.
   • Zero-Shot OOD Test (500 samples): 98.20% Accuracy, 97.20% Attack Recall (catches 243 of 250 novel jailbreak vectors), 99.18% Precision, and 12.04 ms latency.
   • Stage 2 Fact Grounding (648 samples): 75.31% Grounding Accuracy, 78.25% Contradiction Recall (catches 313 of 400 hallucinations), 81.09% Precision, and 16.45 ms latency.
5. Research Paper Deliverables & Visuals: Generated three empirical confusion matrices (`confusion_matrix_v3_in_distribution.png`, `confusion_matrix_v3_zero_shot_ood.png`, `confusion_matrix_v3_factuality.png`), compiled 3-Generation Progression LaTeX Table (`mega_benchmark_results_table.tex`), and updated the Google Colab demonstration suite.

To be filled by Guide (strike off whichever is not applicable)
Performance of students is satisfactory
Performance of students is unsatisfactory
A warning to be issued to student(s) (Name)
Student was not well (Name)
Date 22/09/2026 Signature of Guide